



Faculty : Information Technology
Instructor: Jeremie U. Tuyisenge
Course : Web Technology
Year : 2026-2027

WEB TECHNOLOGY PROJECT INSTRUCTION

You are expected to identify a real-world problem that can be effectively addressed through a web-based software solution. Your proposed project must demonstrate key stages of the software-development life cycle, from requirements and modeling through implementation, testing, and delivery.

1. Provide a concise problem statement, target users, project objectives and scope. Identify principal functional/business requirements and measurable quality attributes (such as usability, performance, reliability and security). Include 2-3 user stories or use cases and acceptance criteria.
2. Domain & Data Modeling:
 - a. Present an ER-style diagram (or UML class diagram) that defines the core entities, their attributes, and the relationships necessary to support your application's data needs.
 - b. Summarize the principal domain concepts (actors, processes, data objects) at an abstract level, independent of implementation details.
 - c. Refine the conceptual model into a technology-oriented schema showing how entities interact within the chosen persistence layer(s). Include primary keys, foreign keys, and relevant constraints or indexes
3. Design and prototype responsive web pages leveraging any modern front-end framework or library (e.g., React, Angular) that address the identified user needs. Mock-ups or wireframes should demonstrate layout adaptability across devices.
4. Implement any architecture styles (monolithic, layered, micro-services, event-driven, etc.) using any framework (e.g., Spring boot , Node.js, Django) to develop the backend. This should facilitate communication with the front end and provide expected outputs to end users
5. Utilize Both Relational (e.g., PostgreSQL, MySQL) and non-relational databases (e.g., MongoDB) to store documents and operation data.
6. Implement secure authentication methods, including OAuth2 mechanisms, and optimize your application for performance.
7. Use RabbitMQ as a message broker for email, SMS, and other events.

8. Define and enforce an authorization model that restricts access to protected resources using RBAC(Role Based Access Control).
9. Use Git throughout the project, with meaningful commits and a clear repository history. Submit a pull/merge request to the main/master branch upon completion.

Bonus point:

1. Integrate DevOps practices into your project, particularly for deployment.
2. Software Testing & Quality Assurance
3. Incorporate any new technology you wish to explore.